

Academic Bio

Kunpeng Yi / 伊坤朋

Associate Researcher / Associate Professor, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences

State Key Laboratory of Regional and Urban Ecology (SKLRUE), formerly State Key Laboratory of Urban and Regional Ecology

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Profile

Research addresses global climate change and eco-environmental remote sensing. Typical ecosystems—wetlands, forests and grasslands—in cities, nature reserves and national parks are examined in relation to drought, flood, wildfire and other extreme meteorological hazards. Threatened migratory birds are treated as indicator species for migration mechanisms, flyway-corridor conservation, and responses to human activity and extreme climate events. Applications include Sino BON, restoration of typical degraded wetlands in Northeast China, bird-habitat planning for the Xiong'an New Area, Nanchang territorial spatial planning (2019–2035), and a bird-strike risk early-warning system. Public academic identity: Movement Ecology × Earth Observation × Biodiversity Conservation.

More than 40 academic papers; more than ten hosted projects (National Natural Science Foundation of China, a National Key R&D; Programme sub-project, and local commissions) with total hosted funding exceeding RMB 8 million; three invention patents; two co-authored or edited books.

Appointments

- Associate Researcher / Associate Professor, Research Center for Eco-Environmental Sciences, Chinese Academy of Sciences; State Key Laboratory of Regional and Urban Ecology (SKLRUE), formerly State Key Laboratory of Urban and Regional Ecology — April 2017 – present
- Postdoctoral Fellow (Ecological Remote Sensing), Aerospace Information Research Institute, Chinese Academy of Sciences (AIRCAS), formerly RADIS — August 2014 – April 2017
- Ph.D., Remote Sensing, Hokkaido University — October 2010 – June 2014. Thesis: 遥感卫星观测及模型在生物质火烧中的应用研究

Selected publications

- Zhang, X., Wan, W., Yang, H. & Yi, K. (2025). Rapid development of wind energy infrastructure threatens bird migration. *The Innovation Geoscience* 3(3): 100142. <https://doi.org/10.59717/j.xinn-geo.2025.100142>
- Tian, Z., Huo, D., Yi, K., Que, J., Lu, Z. & Hou, J. (2024). Evaluation of Suitable Habitats for Birds Based on MaxEnt and Google Earth Engine—A Case Study of Baer's Pochard (*Aythya baeri*) in Baiyangdian, China. *Remote Sensing* 16(1): 64. <https://doi.org/10.3390/rs16010064>
- Yi, K., Meng, F., Gu, D. & Miao, Q. (2023). Optimizing Water Level Management Strategies to Strengthen Reservoir Support for Bird's Migration Network. *Remote Sensing* 15(23): 5508. <https://doi.org/10.3390/rs15235508>
- Yi, K., Zhang, J., Batbayar, N., Higuchi, H., Natsagdorj, T. & Byskatova, I. P. (2022). Using Tracking Data to Identify Gaps in Knowledge and Conservation of the Critically Endangered Siberian Crane (*Leucogeranus leucogeranus*). *Remote Sensing* 14(20): 5101. <https://doi.org/10.3390/rs14205101>

Students and service

Council member, China Grassland Society. Member of the Geographical Society of China, the Ecological Society of China, the Chinese Society of Remote Sensing, and the China Ornithological Society. Guest Editor of special issues for *Remote Sensing*, *Land and Fire*.

- Master's supervisor (硕导).
- Directions: 运动生态学, 迁徙动物监测与保护; 湿地监测与生态修复; 生态系统遥感监测与评估.

- Majors: 071300 Ecology; 0705Z1 Natural Resources Science; 070503 Cartography and Geographic Information Systems.
- Prospective students: write to kpyi@rcees.ac.cn.